

SEONGTAE BANG

GPU-Efficient Heterogeneous Systems · Efficient AI Training Systems · GPU Memory Systems

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RESEARCH PROFILE

Ph.D. researcher in computer architecture and systems focused on **GPU-efficient heterogeneous systems for memory-intensive AI workloads**.

I identify computation and state that do not need to occupy GPU resources, then restructure execution so that they progress concurrently on CPUs or other system resources. This frees HBM capacity and GPU cycles for highly parallel, accelerator-critical work, improving memory efficiency, GPU utilization, and end-to-end throughput.

My work spans multi-GPU training frameworks, NVIDIA GPU system software, and GPU architecture research. I have modified DeepSpeed and Megatron-LM, redesigned NVIDIA UVM mechanisms at the driver level, and proposed new GPU memory-management architectures through simulation, with first-author papers at **ASPLOS 2027** and **MICRO 2026**.

SELECTED RESEARCH & ENGINEERING

ReFlow — GPU-Efficient LLM Training through CPU-GPU Parallelism First author · ASPLOS 2027 (*to appear*)

- Identified the exposed host update in CPU-offloaded training as a host-memory-traffic problem rather than a CPU-arithmetic one: non-arithmetic overhead is about **5×** the Adam compute time, and the GPU idles for up to **47.9%** of each iteration on a B200 system.
- Redesigned the optimizer-offload path of DeepSpeed-based LLM training to decouple CPU optimizer work from accelerator-critical forward and backward execution, implemented end-to-end across the framework runtime, gradient transfer, CPU optimizer kernels, and multi-GPU execution.
- Overlaps CPU optimizer work with GPU training via register-resident SIMD fusion, BF16 gradient transfer with CPU-side accumulation, and deadline-aware scheduling, preserving exact training semantics (bit-identical losses).
- Reduces GPU memory consumption while achieving throughput comparable to or higher than GPU-resident ZeRO-3 across evaluated configurations (**98%** at 50B on 8 B200 GPUs; **1.2×** higher at 80B). Achieves up to **4×** the throughput of ZeRO-Infinity and **3×** that of SuperOffload, and scales to **95B** parameters on 8 B200 GPUs (ZeRO-Infinity: 80B) with about 20% lower host-memory footprint.
- Builds on ReplayOpt (IEEE CAL 2026, first author), which cut CPU optimizer time by up to **55.2%** and training-step time by up to **21.7%**.
- Artifact publicly available as a DeepSpeed fork (Apache-2.0); planning to contribute ReFlow upstream to DeepSpeed through a pull request.

ReclaimX — GPU Architecture Redesign for UVM Oversubscription First author · MICRO 2026 (*to appear*)

- Analyzed GPU execution under UVM oversubscription and identified a previously unused opportunity: fault-stalled execution resources cannot advance foreground work during page-fault recovery.
- Proposed repurposing those resources for device-side memory reclamation, combining fine-grained victim prioritization, selective compressed-resident retention, and local restoration to avoid unnecessary host migration.
- Validated with GPU simulation and UVM modeling: **2.33×** geometric-mean speedup over baseline UVM across 12 oversubscribed workloads; with prefetching, **90.6%** fewer far-page faults and up to **3.61×**.

SAFE & ARIADNE — NVIDIA UVM Driver Redesign Co-first author · CAL 2025 / Second author · HPCA 2026

- Performed code-level analysis of NVIDIA's open-source UVM driver (page faults, migration, prefetching, sharing, oversubscription policies) and modified it directly rather than treating it as a black box.
- SAFE: redesigned prefetching decisions using workload sharing and access behavior to reduce ineffective prefetching and unnecessary HBM traffic; up to **6.5×** (**3.6×** average) over the default UVM prefetcher.
- ARIADNE: redesigned adaptive UVM management to respond to runtime memory demand under oversubscription; **1.9×**, **5.0×**, and **4.8×** average speedups over a prior state-of-the-art method at 130%, 175%, and 300% oversubscription.

Memory-Scalable MoE Training Ongoing

- Extending Megatron-LM with heterogeneous CPU-GPU execution that offloads optimizer and state-management work not requiring GPU resources, while keeping grouped expert GEMMs on GPUs.
- Aims to free HBM so that TP/PP/DP configurations are chosen for throughput rather than memory feasibility, increasing trainable model capacity.

Memory-Scalable 3D Gaussian Splatting Training Ongoing

- Pipelining CPU-side culling for upcoming views with current-view GPU rasterization and backward propagation.
- Keeps only the active Gaussian working set in HBM and persistent training state in host memory, aiming to train beyond practical HBM capacity without changing the training objective.

SYSTEMS DEVELOPMENT

DeepSpeed — Modified LLM training and optimizer-offload paths (optimizer execution, CPU–GPU synchronization, gradient movement, mixed precision); evaluated GPU-resident vs. CPU-offloaded training on multi-GPU systems.

Megatron-LM — Modified MoE training paths involving expert execution, optimizer/state placement, GPU residency, and CPU offloading; analyzed interactions among TP, PP, DP, EP, HBM capacity, and training throughput.

NVIDIA UVM / GPU memory software — Source-level analysis and modification of page-fault handling, migration, prefetching, and oversubscription; architecture simulation for mechanisms beyond current hardware and drivers.

TECHNICAL EXPERTISE

Languages	C · C++ · Python · CUDA · Bash
AI Training Systems	DeepSpeed · Megatron-LM · PyTorch · LLM Training · Mixture-of-Experts · TP/PP/DP/EP · ZeRO · Optimizer Offload · Mixed Precision · Grouped GEMM
GPU Systems & Drivers	NVIDIA UVM · GPU Driver Internals · Linux Kernel Drivers · Page-Fault Handling · Page Migration · Prefetching · GPU Memory Oversubscription · HBM Working-Set Management · Memory Tiering · NVIDIA nvCOMP
Heterogeneous & CPU	CPU–GPU Co-execution · Workload Partitioning · Overlap and Scheduling · AVX-512 SIMD · OpenMP · NUMA · Multithreaded Optimization · Memory-Layout Optimization
Performance Analysis Simulation	Nsight Systems · Nsight Compute · perf · Workload Characterization · Memory-Traffic Analysis · Accel-Sim · UVMsmart · gem5 · pNet-gem5 · GPU Memory-System Simulation
Platforms	NVIDIA B200 · A100 · RTX PRO / RTX A-series · AMD EPYC Multi-Socket Systems · Intel IAA

PUBLICATIONS

ReFlow: Exposing Parallelism in CPU-Offloaded LLM Training via Register-Resident SIMD Fusion and Decoupled Update Scheduling. Seongtae Bang, Gyeongseo Park, Ki-Dong Kang, Hyunkyun Shin, Sungju Kim, and Daehoon Kim. *Proceedings of the 32nd ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '27)*, Heraklion, Greece, April 2027 [To appear](#)

ReclaimX: Device-Side Memory Reclamation via Stalled GPU Execution for UVM Oversubscription. Seongtae Bang, Hyunkyun Shin, Hyungwon Park, Minho Kim, and Daehoon Kim. *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2026 [To appear](#)

ARIADNE: Adaptive UVM Management for Efficient GPU Memory Oversubscription. Hyunkyun Shin, Seongtae Bang, Hyungwon Park, and Daehoon Kim. *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2026

ReplayOpt: Optimizer-State Replay to Resolve Critical-Path Bottlenecks in Offloaded Training. Seongtae Bang, Gyeongseo Park, Kyeonghyeon Ryu, and Daehoon Kim. *IEEE Computer Architecture Letters*, 2026

SAFE: Sharing-Aware Prefetching for Efficient GPU Memory Management with Unified Virtual Memory. Hyunkyun Shin*, Seongtae Bang*, Hyungwon Park, and Daehoon Kim. *IEEE Computer Architecture Letters*, 2025

pNet-gem5: Full-System Simulation with High-Performance Networking Enabled by Parallel Network Packet Processing. Jongmin Shin, Seongtae Bang, Gyeongseo Park, and Daehoon Kim. *IEEE Computer Architecture Letters*, 2025

* Equal contribution.

RESEARCH EXPERIENCE

Graduate Researcher — CASLAB 2020–Present

- Research across AI training systems, GPU memory systems, heterogeneous computing, and computer architecture: redesign runtime mechanisms in software, and use architecture simulation when hardware changes are needed.
- Develop and modify established frameworks (DeepSpeed, Megatron-LM) rather than standalone prototypes, and study how lower persistent HBM requirements enable higher-throughput TP/PP/DP/EP configurations.
- Build optimized CPU paths (AVX-512 SIMD, OpenMP, NUMA-aware parallelism, memory-layout optimization), evaluate Intel IAA and NVIDIA nvCOMP compression trade-offs, and characterize systems end-to-end across GPU compute, CPU execution, HBM capacity, host-memory bandwidth, and data movement.

EDUCATION

DGIST — Ph.D. Student (Integrated M.S.–Ph.D.), Electrical Engineering and Computer Science 2020–Present
Advisor: Prof. Daehoon Kim · Expected Graduation: February 2027

DGIST — Bachelor of Convergence Engineering 2016–2020